The **FAULT** Framework for **Failure Analysis** and Understanding for Language-to-SPARQL Testing

Indeewari Balasooriya · **Bianca Benelli** · **Niccolò Marastoni** · **Elisa Quintarelli** · **Matteo Lissandrini**

University of Verona, Verona, Italy · name.surname@univr.it

Why Aggregate Accuracy Isn't Enough to Compare NL2SPARQL Translation Systems?

A single score hides *why* a system fails:

- **Noisy question phrasing**
- **SPARQL structural complexity**
- **Schema-grounding errors**
- **Lexicalized IRI shortcuts**

FAULT: The Idea

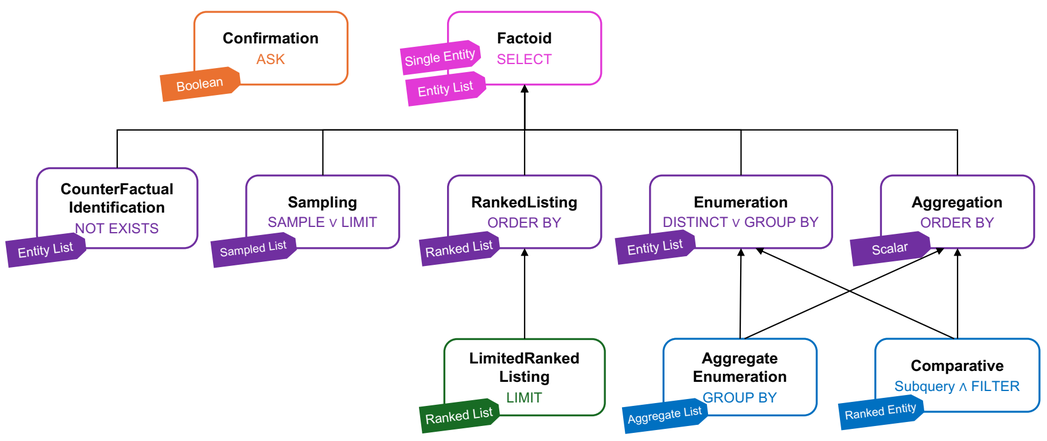
A **reusable annotation methodology** + an **open-source toolchain** layered onto existing NL2SPARQL benchmarks, ***not a new benchmark.***

FAULT adds **three diagnostic annotations:**

- **Question-type labels**
- **Structural features**
- **Noise annotation**

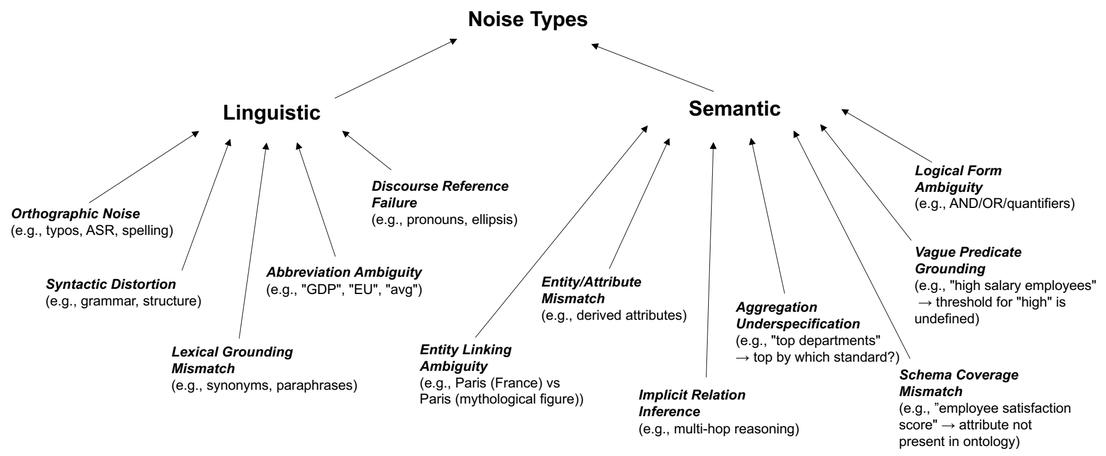
Axis 1 · Question Type

Grounded in **SPARQL operators**, not phrasing



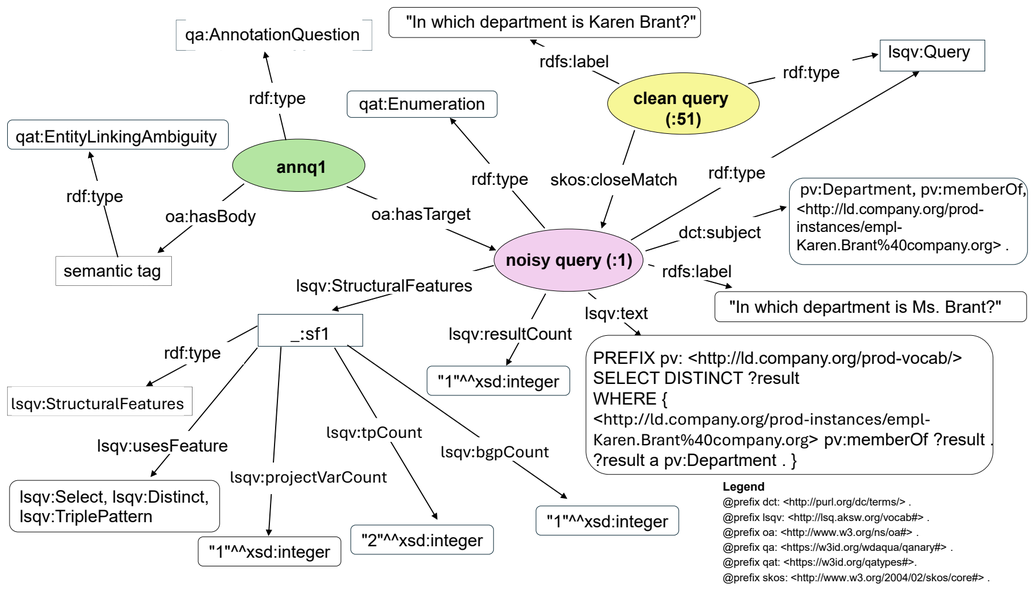
Axis 2 · Noise Taxonomy

Phrasing failures vs. **schema-resolution** failures



† Intrinsically unanswerable: excluded from the practical performance ceiling.
Manually annotated; disagreements resolved by voting.

Annotation Graph Example



The **noisy question** is ambiguous ("Ms. Brant" matches >1 employee) and linked via *skos:closeMatch* to its disambiguated **clean counterpart**, which carries the reference structural features and question-type label.

Annotation Workflow

- **Parse** query → annotate question type and extract LSQ features
- **Flag** anti-patterns automatically
- **Annotate** noise; resolve by voting
- **Link** noisy ↔ clean via closeMatch

216
annotated queries

2
benchmarks
95 CK25 + 121 DBpedia

IRI Obfuscation

Readable IRIs (e.g. *pv:Department*) let LLMs "guess" without real schema understanding. FAULT **hashes local names, namespace-aware:**

hash = SHA-256(local_name) mod 10^6

- Labels & literals **untouched**
- Namespace prefix **kept**

Evaluation: SPARQL-LLM

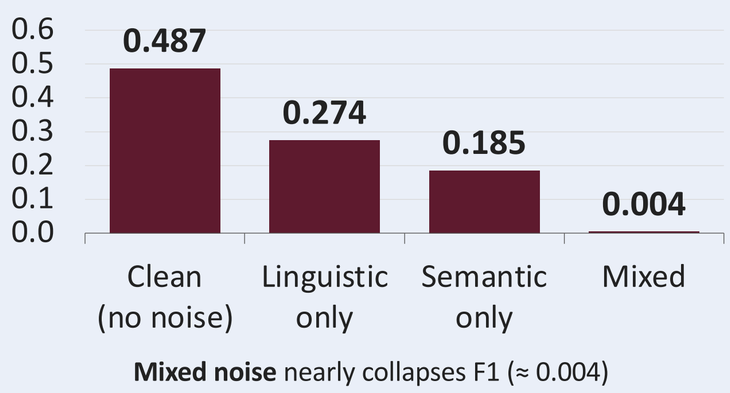
Retrieval-augmented system (schema + query exemplars) on **CK25** (50 Q) and **DBpedia** (100 Q), Text2SPARQL'25. **F1** over result bindings.

	CK25	DBpedia
Classes	20	752
Predicates	50	61.9K
Triples	27K	173.7M

- Original vs. hashed IRIs · with vs. without examples
- CK25 noisy Qs: **16** linguistic / **23** semantic / **6** mixed

Noise-Type Effect

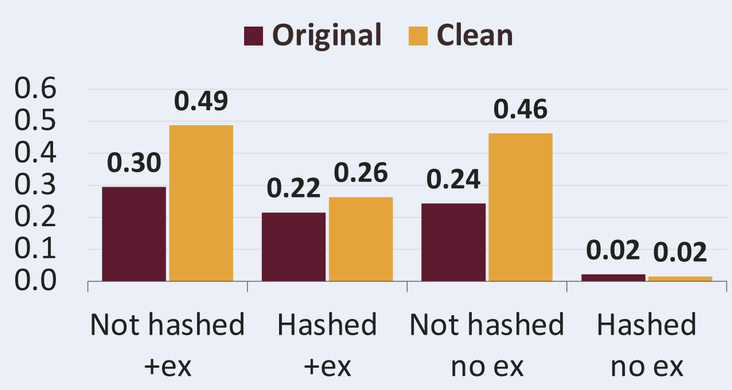
CK25, best config (+examples, not hashed)



Mixed noise nearly collapses F1 (≈ 0.004)

IRI-Hashing Effect

F1 on CK25: original vs. clean queries



Key Findings

F1: 0.487 → 0.263
It **drops** when IRIs are hashed

F1: 0.243 → 0.295
Examples help, even without legible IRIs

F1: 0.274 / 0.185 / 0.004
Linguistic / semantic / mixed noise all lower F1

Limitations & Roadmap

Validated on **2 datasets, 1 system:** a scoping choice, not a claim of generality

Port FAULT to QALD, LC-QuAD 2.0, SciQA, DBLP-QuAD, QueryBridge

Zero-noise GPTKB benchmark, adding schema complexity as a new dimension

This work has been supported by **DataGEMS**, funded by the European Union's Horizon Europe Research and Innovation program, under grant agreement No 101188416

DataGEMS

Funded by the European Union

Resources
Public under CC BY 4.0

Ontology · **SPARQL Annotator** · **Annotation Tool** · **Data & Scripts** · **Archived Release**